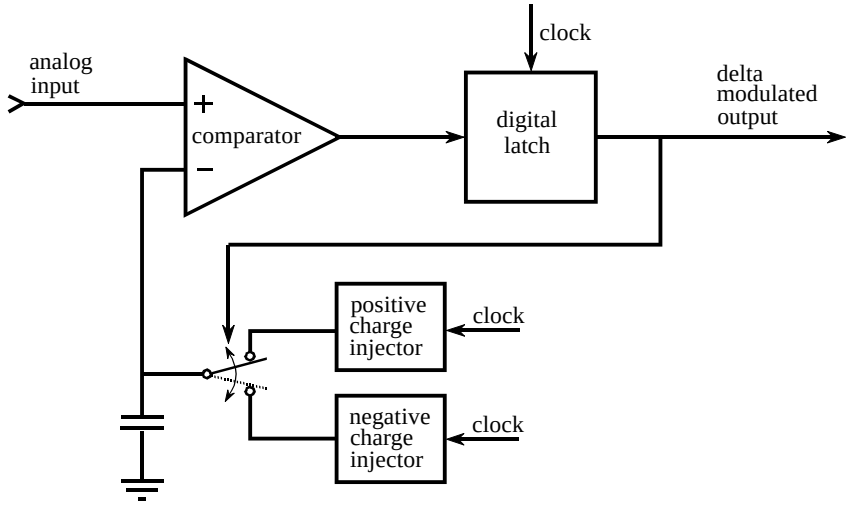

FIGURE 3-16
Block diagram of a delta modulation circuit. The input voltage is compared with the voltage stored on the capacitor, resulting in a digital zero or one being applied to the input of the latch. The output of the latch is updated in synchronization with the clock, and used in a feedback loop to cause the capacitor voltage to track the input voltage.

input, to its output. This latch insures that the output is synchronized with the clock, thereby defining the sampling rate, i.e., the rate at which the 1 bit output can update itself.

A feedback loop is formed by taking the digital output and using it to drive an electronic switch. If the output is a digital *one*, the switch connects the capacitor to a *positive charge injector*. This is a very loose term for a circuit that increases the voltage on the capacitor by a fixed amount, say 1 millivolt per clock cycle. This may be nothing more than a resistor connected to a large positive voltage. If the output is a digital *zero*, the switch is connected to a *negative charge injector*. This *decreases* the voltage on the capacitor by the same fixed amount.

Figure 3-17 illustrates the signals produced by this circuit. At time equal zero, the analog input and the voltage on the capacitor both start with a voltage of zero. As shown in (a), the input signal suddenly increases to 9.5 volts on the eighth clock cycle. Since the input signal is now more positive than the voltage on the capacitor, the digital output changes to a *one*, as shown in (b). This results in the switch being connected to the positive charge injector, and the voltage on the capacitor increasing by a small amount on each clock cycle. Although an increment of 1 volt per clock cycle is shown in (a), this is only for illustration, and a value of 1 millivolt is more typical. This staircase increase in the capacitor voltage continues until it exceeds the voltage of the input signal. Here the system reached an equilibrium with the output oscillating between a digital one and zero, causing the voltage on the capacitor to oscillate between 9 volts and 10